Heat Flow Across a Material Interface

Conservative FDM and piecewise-linear FEM

MTH321 Project 2

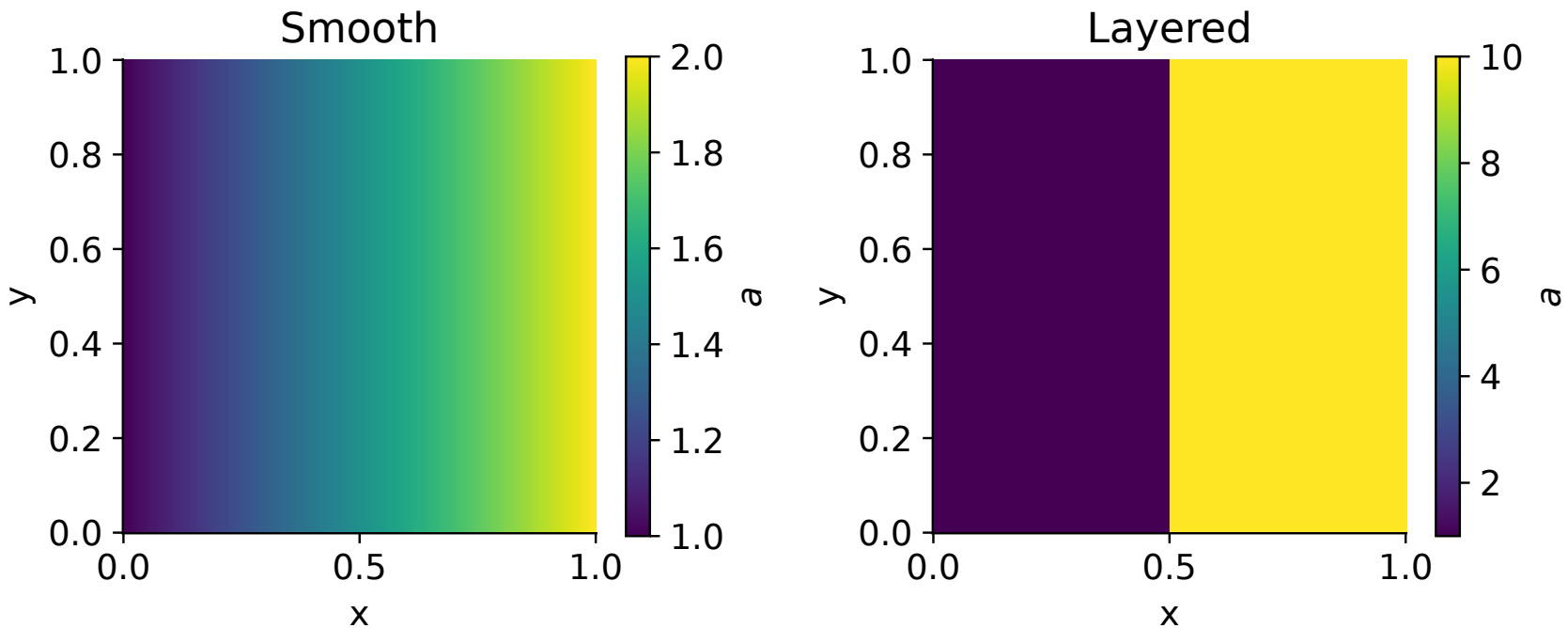
XJTLU

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What must a numerical method preserve when conductivity jumps?

Two conductivity models

$$-\nabla\cdot(a\nabla u) = f, \qquad \mathbf{q} = -a\nabla u, \qquad u = g \text{ on } \partial\Omega$$



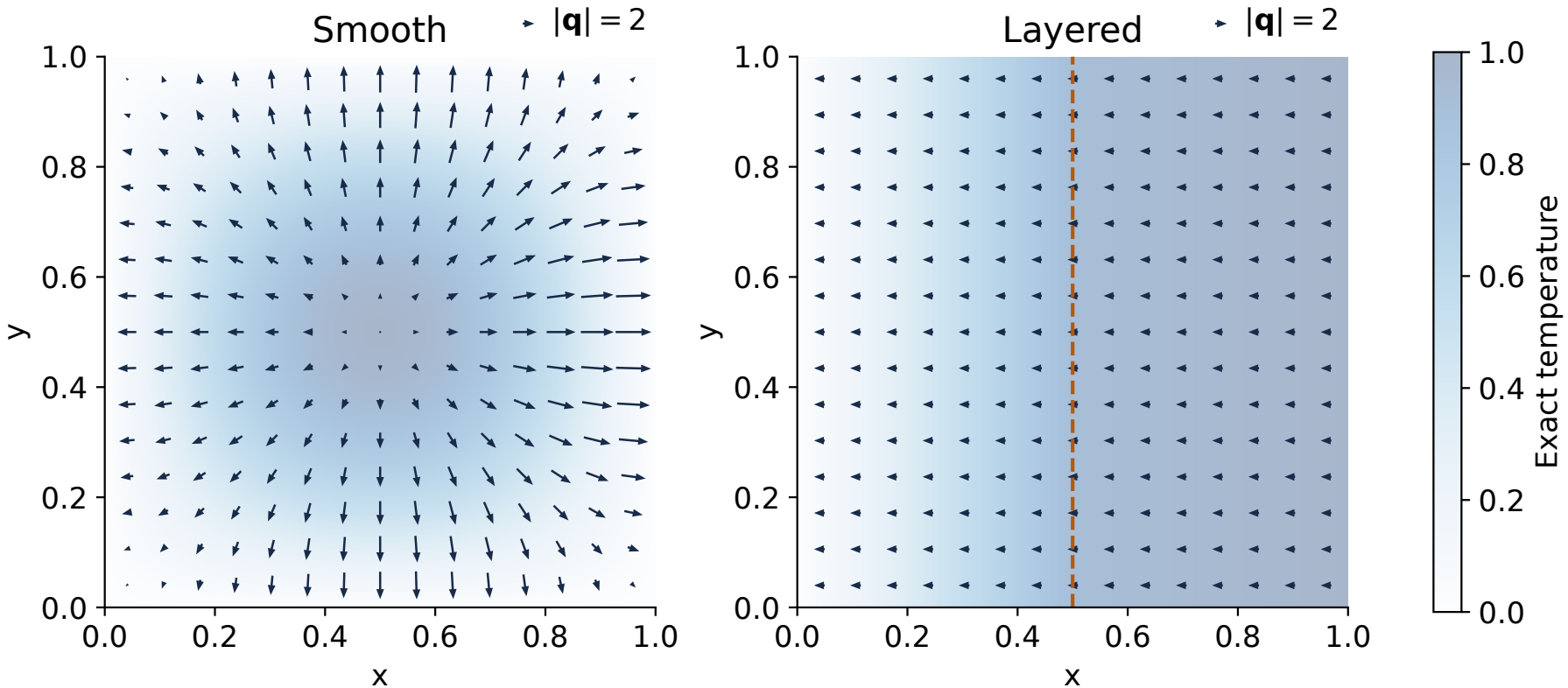
Smooth: accuracy

$a = 1 + x$, $u^* = \sin(\pi x)\sin(\pi y)$.

Layered: interface physics

$a_L = 1$, $a_R = 10$; interface $x = 1/2$.

Heat flux and the interface conditions

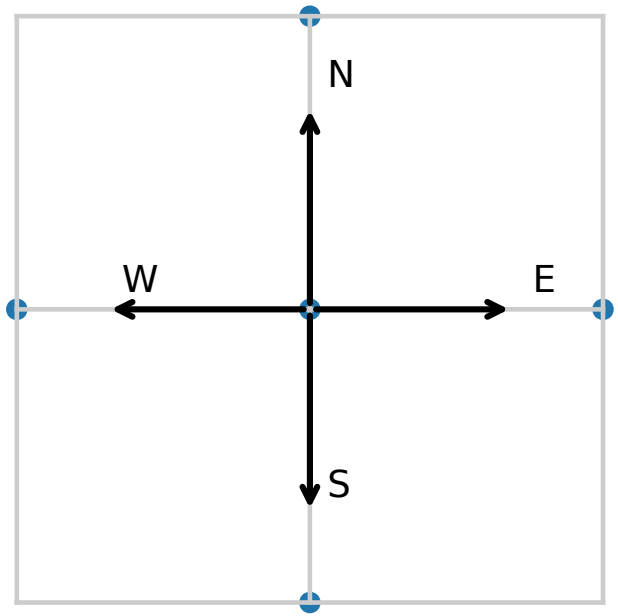


$$[u] = 0, \qquad [\mathbf{q} \cdot \mathbf{n}] = 0, \qquad a_L\, u_x^- = a_R\, u_x^+.$$

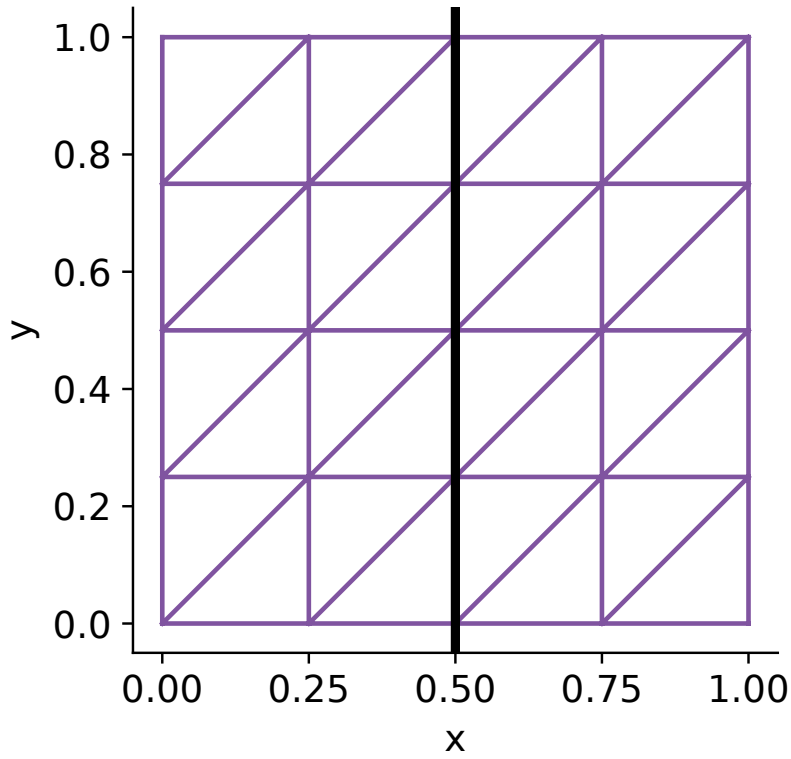
The low-conductivity material needs a steeper temperature gradient.

Two numerical representations

FDM: shared face fluxes



FEM: interface-fitted triangles



FDM: balance shared fluxes

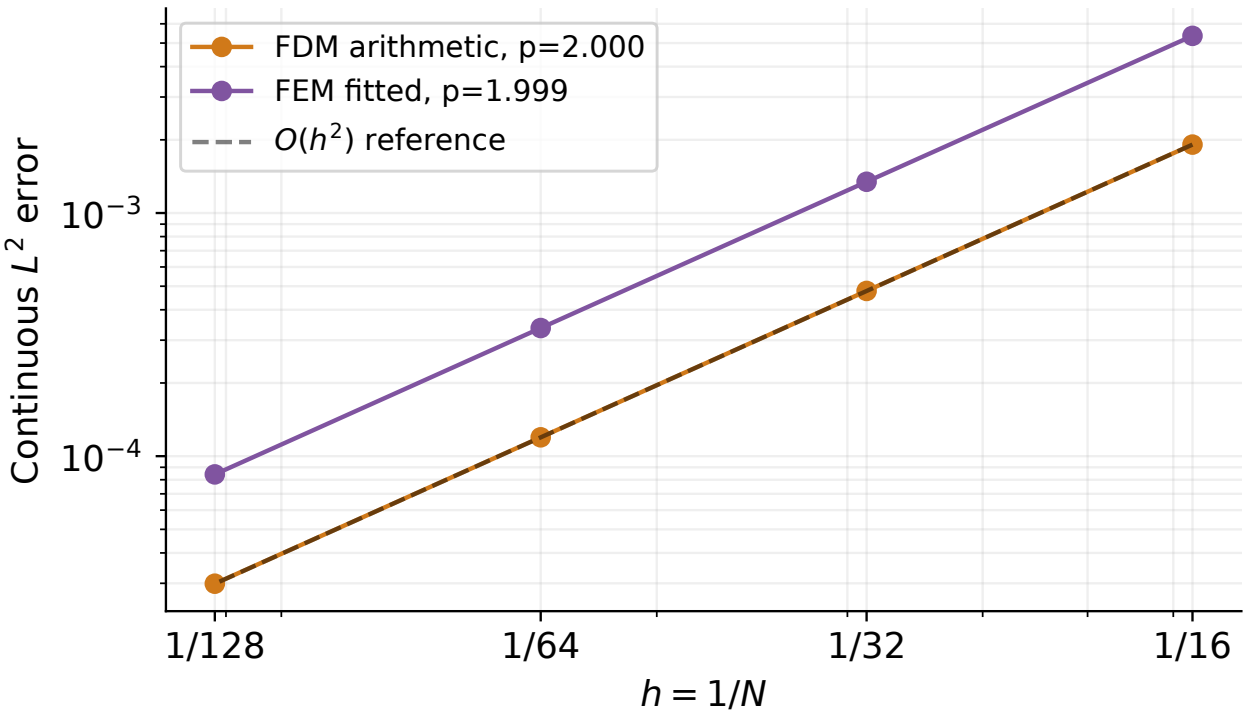
$$(A_h U)_P = \frac{1}{h^2} \sum_s a_s (U_P - U_s).$$

FEM: satisfy weak balance

$$\int_\Omega a \nabla u_h \cdot \nabla v_h = \int_\Omega f v_h.$$

Dirichlet unknowns are eliminated; both reduced systems are SPD.

Smooth benchmark: second-order accuracy



Prediction

$E_{L^2} \sim Ch^2$.

Fitted orders

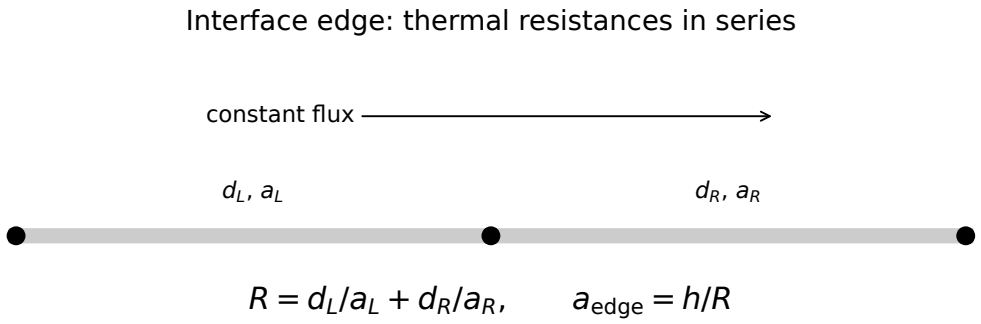
FDM: 1.9999

FEM: 1.9990

Four meshes; common continuous L^2 norm.

Algebraic and quadrature effects are below the measured error.

Representing the material interface



$$a_{\text{edge}} = \frac{h}{d_L/a_L + d_R/a_R}.$$

Bisected edge: $a_H = \dfrac{2a_L a_R}{a_L + a_R}$.

Fitted FEM

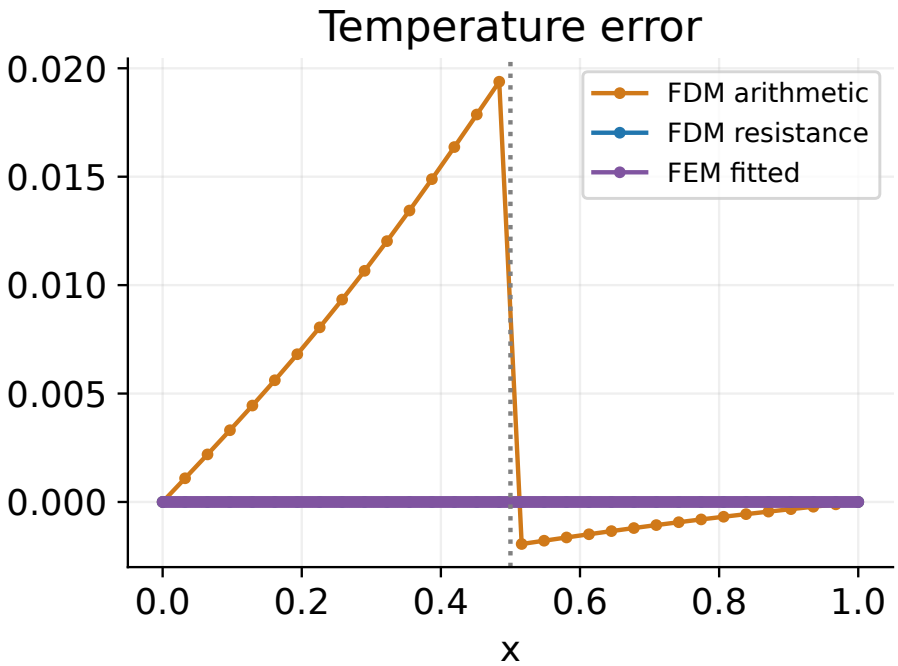
The exact layered solution is continuous and linear in each material.

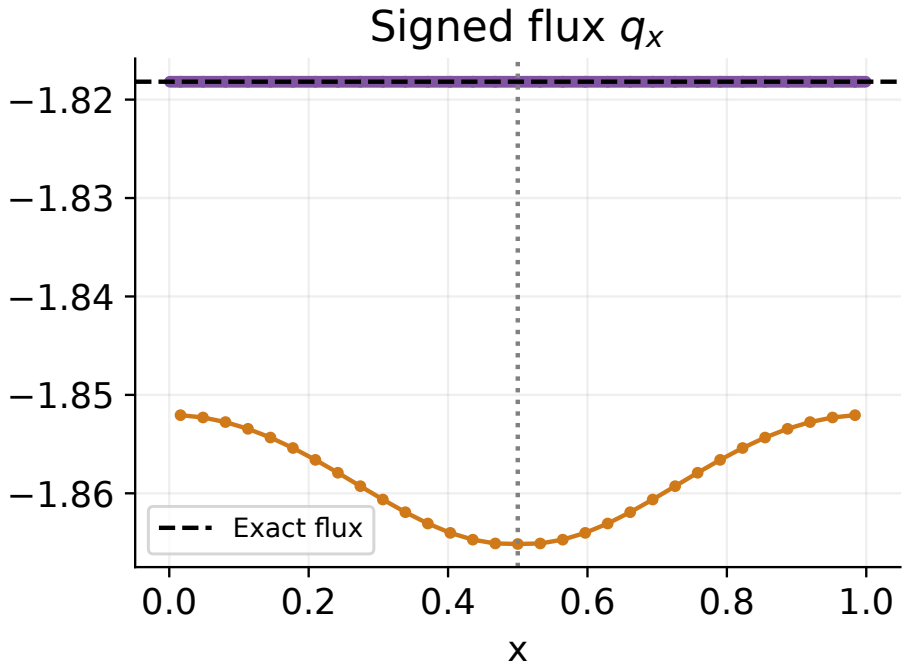
With the interface on element edges, $u^* \in V_h$.

Galerkin uniqueness gives $u_h = u^*$, up to numerical error.

Correct resistance in FDM; an exactly representable kink in FEM.

Layered benchmark: temperature and flux





- Resistance FDM and fitted FEM recover the exact benchmark.
- Arithmetic FDM remains bounded and symmetric, yet gives wrong flux.

$N = 31$; slices: FDM $y = 0.484$, FEM $y = 0.506$.

Refinement and contrast increase solver work

Unpreconditioned CG: refining $N = 15 \rightarrow 63$

Contrast a_R/a_L	Resistance FDM	Fitted FEM
1	$44 \rightarrow 197$	$53 \rightarrow 221$
10	$90 \rightarrow 470$	$94 \rightarrow 472$
100	$168 \rightarrow 1190$	$189 \rightarrow 1205$

Same stopping rule: $\|b - AU\|_2 / \|b\|_2 \leq 10^{-10}$.

- Refinement and higher contrast increase work in the tested cases.
- Correct interface physics does not remove conditioning difficulties.

What the experiments establish

1. **Smooth accuracy:** both methods approach second order in L^2.
2. **Interface physics:** resistance FDM and fitted FEM reproduce this layered benchmark to numerical precision.
3. **Validation:** symmetry and bounded temperature are insufficient; inspect the heat flux as well.

Representing the material interface is part of representing the PDE.